8 **(a)** In an α-particle scattering experiment, an α-particle is travelling in a vacuum towards the centre of a gold nucleus, as illustrated in Fig. 8.1.

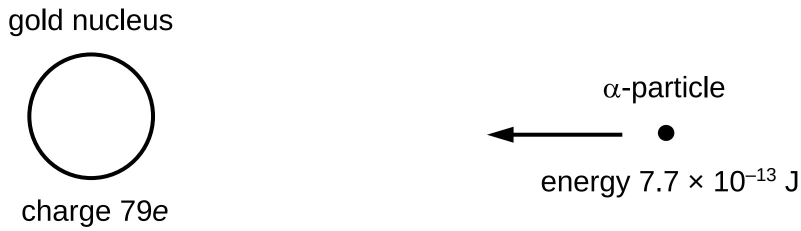

Fig. 8.1

The gold nucleus has a charge 79*e*. At a large distance from the gold nucleus, the α-particle has energy 7.7×10^{-13} J.

(i) The α-particle does not collide with the gold nucleus.

Show that the radius of the gold nucleus must be less than 4.7×10^{-14} m.

[2]

(ii) Fig. 8.2 shows three α-particles, all with the same kinetic energy and travelling in the same initial direction, as they approach a stationary gold nucleus.
The path followed by one of the α-particles is shown.

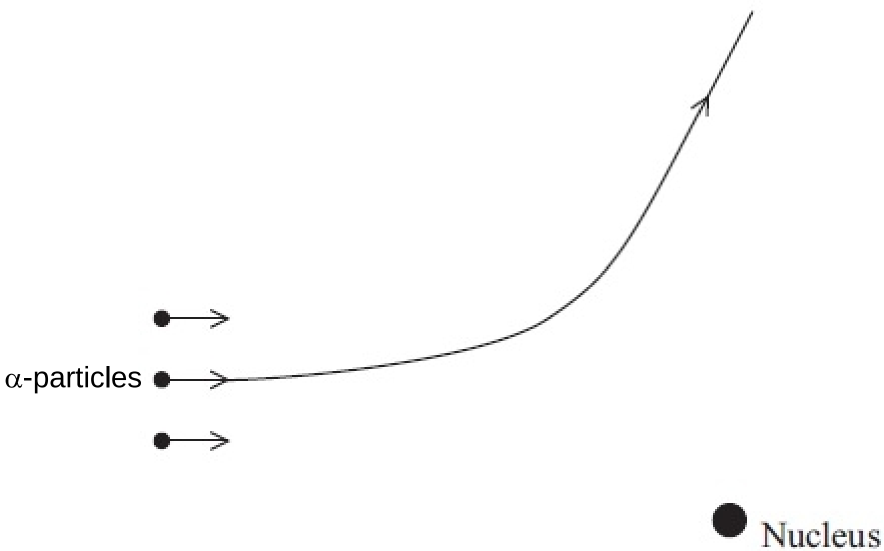

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Fig. 8.2 (not to scale)

On Fig. 8.2, complete the paths of the other two α-particles as they approach and pass by the nucleus. [2]

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(b) Some data for nuclei are given in Fig. 8.3.

Particle/ Nuclide	**Name**	**Mass / u**	**Binding energy per nucleon / MeV**
$^{1}_{1}H$	Proton	1.00728	-
$^{1}_{0}n$	Neutron	1.00866	-
$^{4}_{2}He$	Helium		7.07470
$^{14}_{7}N$	Nitrogen		7.47724
$^{17}_{8}O$	Oxygen		7.75224

Fig. 8.3

(i) Explain what is meant by *binding energy per nucleon* for the helium nucleus.

....

....

.... [2]

(ii) Use data from Fig. 8.3 to determine the mass of helium nucleus.

mass = ... u [2]

(iii) In a particular nuclear reaction, a slow moving alpha particle bombards a stationary nitrogen nucleus ${}^{14}_{7}\text{N}$, transmuting it to an oxygen nucleus ${}^{17}_{8}\text{O}$ and a proton.

The nuclear reaction is shown below.

$${}^{14}_{7}\text{N} + {}^{4}_{2}\text{He} \rightarrow {}^{17}_{8}\text{O} + {}^{1}_{1}\text{H}$$

Use data from Fig. 8.3 to determine whether the reaction can occur spontaneously. Explain your working.

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[3]

(c) A radiation detector is placed close to a radioactive source.

The variation with time t of the measured count rate is shown in Fig. 8.4.

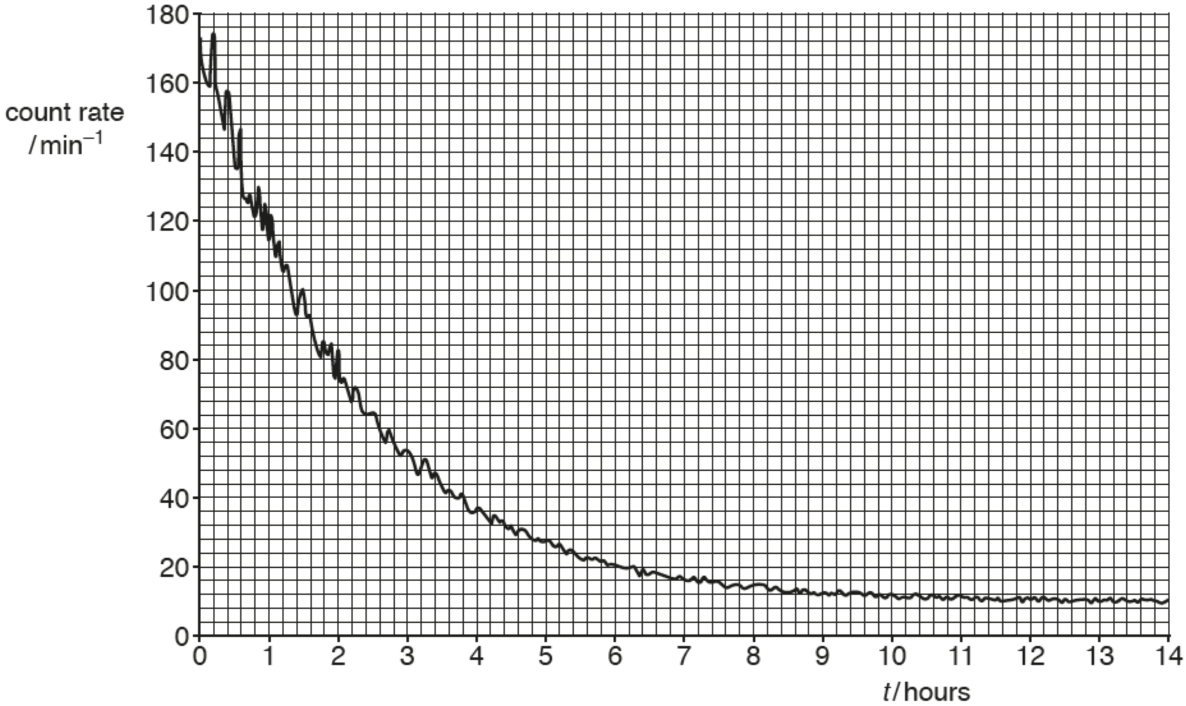

Fig. 8.4

The graph is used to determine the half-life of the radioactive source.

(i) With reference to the graph, explain how we can deal with the problem of the random nature of count rate.

....

.... [1]

(ii) With reference to the graph, explain how we can deal with the problem of background radiation.

....

.... [1]

(iii) Hence use Fig. 8.4 to determine the half-life of the radioactive source.

half-life = ... hours [2]

(iv) The readings in Fig. 8.4 were obtained at room temperature.

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A second sample of this radioactive source is heated to a temperature of 500 °C.
The initial count rate at time $t = 0$ *is* the same as that in Fig. 8.4.
The variation with time t of the measured count rate from the heated source is determined.

State and explain if there are any differences for the 2 samples in

1. the half-life,

....

.... [1]

2. the measured count rate at any time *t.*

....

.... [1]

(d) A small volume of solution containing the radioactive isotope sodium-24 ($^{24}_{11}Na$) has an initial activity of 3.8×10^4 Bq.

Sodium-24 has a half-life of 15 hours and decays to form a stable daughter isotope.

All of the solution is poured into a container of water. After 36 hours, a sample of water of volume $5.0\ cm^3$, taken from the container, is found to have an activity of 1.2 Bq.

Assuming that the solution of the radioactive isotope is distributed uniformly throughout the container of water, calculate the volume of water in the container initially.

volume = ... cm^3 [3]

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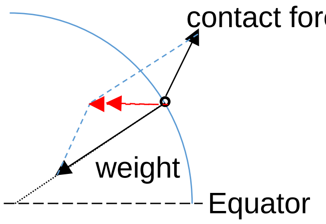
Answers and Marking Scheme

1	(a)	(i)	 • Weight pointing towards the centre of the Earth • Contact force by ground on person comprises of normal contact force and friction. Accept if normal contact force and friction force are drawn separately. • Resultant force parallel to the plane of equator and pointing towards the axis of rotation. • Wrong labelling -1	B1 B1
		(ii)	• At the poles there is no rotation, hence the centripetal force required is zero. • The weighing scale reads the normal contact force which is equal to the true weight of the boy at the North pole. Thus scale reading is larger than the reading at the current location.	B1 B1
	(b)	(i)	Gravitational force provides the centripetal force for circular motion $\frac{GMm}{r^2} = m\omega^2 r = m\left(\frac{4\pi^2}{T^2}\right)r$	C1

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			$r^3 = GM\dfrac{T^2}{4\pi^2} = \left(6.67\times10^{-11}\right)\left(6.02\times10^{24}\right)\dfrac{\left(24\times3600\right)^2}{4\pi^2}$ $r = 4.23\times10^7$ m Height above the boy or surface of Earth (boy's height is negligible) $h = 4.23\times10^7 - 6400\times10^3 = 3.59\times10^7$ m	A1 A1
		(iii)	It is not a geostationary satellite as it does not orbit in the same plane as the equator.	B1

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2	**(a)**	ΔU : Increase in internal energy Q : heat supplied to the system W : Work done on system	B1
	(b)(i)	$Q = ml_v = 0.37\times2.3\times10^6 = 8.51\times10^5$ J	A1
	(b)(ii)	No of moles $n = \dfrac{M}{M_R} = \dfrac{370}{18} = 20.6$ Using $PV = nRT$ $\Rightarrow 1.0\times10^5 V = 20.6\times8.31\times(273+100)$ $\Rightarrow V = 0.64$ m^3	C1 M1 A0
	(b)(iii)	Work done by water = $P\Delta V = 1.0\times10^5\times0.64$ = 6.4×10^4 J	C1 A1
	(b)(iv)	Work done on water, $W = -6.4\times10^4$ J By 1st law, $\Delta U = Q + W = 8.51\times10^5 - 6.4\times10^4 = 7.9\times10^5$ J	C1 A1
	(b)(iv)	Increase in internal energy is due to the increase in potential energy of the molecules when the separation between the molecules increases as it vaporizes. There is no increase in the kinetic energy of the molecules since kinetic energy of molecules is proportional to thermodynamic temperature and there is no change in temperature.	B1 B1

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3(a) At equilibrium,

$T_1 = ke = mgsin\vartheta$

$T - mgsin\vartheta = ma$ M1

$T - T_1 = ma$ A1

$k(x+e) - ke = ma$ (Newton's Second Law)

Since x is measured downwards from equilibrium, A0

$$a = -\frac{kx}{m}$$

(b)

$$v = \pm\omega\sqrt{x_o^2 - x^2}$$

$$v_o = \omega x_o$$

$$\frac{v}{v_o} = \frac{\sqrt{x_o^2 - x^2}}{x_o}$$

M1

$$0.25 = \frac{\sqrt{3^2 - x^2}}{3}$$

$$x = 2.9 \text{ cm}$$

A1

Or

$v = \pm\omega\sqrt{(x_o^2 - x^2)} = 0.25\ \omega x_o$

$x_o^2 - x^2 = 0.0625\ x_o^2$

x = 2.9 cm

(c)(i) Resonance is illustrated. The driving frequency of the oscillator is equal to the natural frequency of the mass-spring system, maximum energy is transferred to the system and the mass-spring system oscillates with maximum amplitude. B1 B1

(c)(ii) Magnitude of acceleration = $\omega^2 x_o$

$= [2\pi(1.2)]^2(0.06)$ M1

$= 3.4\ ms^{-1}$ A1

4 (a) It states that the electrostatic force of interaction between 2 point charges is proportional to the product of the 2 charges and inversely proportional to the square of the distance between them. B1

(b)(i)

$$E = \frac{1}{4\pi\varepsilon_o}\frac{Q}{r^2}$$

$$E_A = E_X - E_Y$$

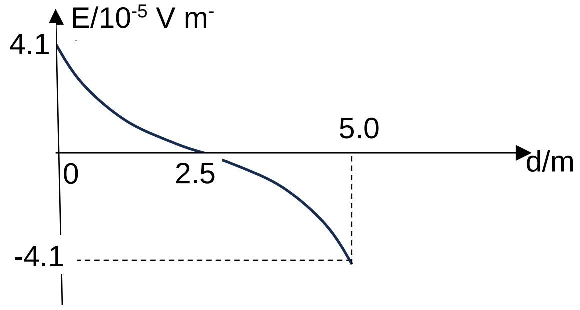
C1

$$4.1\times10^{-5}=\frac{Q}{4\pi\times8.85\times10^{-12}\times0.025^2}-\frac{Q}{4\pi\times8.85\times10^{-12}\times0.075^2}$$ M1 A1

$\Rightarrow$ Q = 3.2 × 10^{-18} C

(b)(ii) Smooth curve with gradient decreasing starting at (0, 4.1 × 10^{-5}) to d-axis at (2.5, 0). B1

Smooth curve with gradient increasing from (2.5, 0) ending at (5, - 4.1 × 10^{-5}). B1



(b)(iii) Change in potential / potential difference between AB B1

(b)(iv) Acceleration decreases to zero at midpoint; B1

1. Then acceleration decreases in the opposite direction/ increasing negative acceleration B1

2. Oscillation / oscillatory motion B1

5 **(a)(i)** Heating effect in a resistor $\propto$ (current)^2 B1

so square of value of alternating current is always positive B1

OR (current moves in opposite direction in resistor during half-cycles. (B1)

But heating effect is independent of direction) (B1)

(a)(ii) The value of the direct current that produces the same heating effect or power (as the alternating current) in a resistor B1

(b)(i) Period $T=\frac{1}{f}=\frac{1}{50}=0.020$ s C1

Hence, t = 0.030 s A1

(b)(ii) Peak voltage, $V_o=17.0$ V C1

$V_{rms}=\frac{V_o}{\sqrt{2}}=\frac{17.0}{\sqrt{2}}=12.0$ V A1

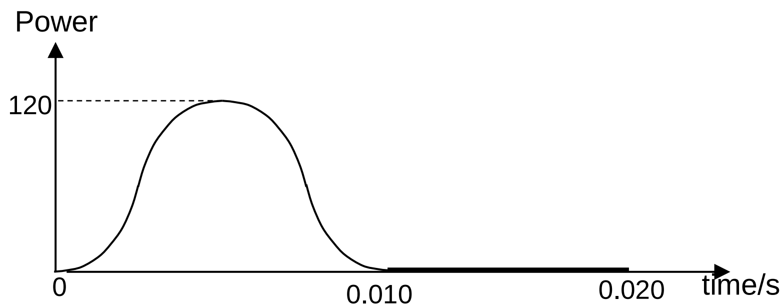
(b)(iii) Sine square curve for half a cycle with maximum power of ≈120 W and a horizontal straight line in the 2nd half cycle

or vice-versa

correct shape B1

maximum at ≈120 W B1

initial zero gradient for sine square graph B1



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6 **(a)** **(i)** threshold frequency = 1.00×10^{15} Hz (allow $\pm 0.05 \times 10^{15}$ Hz) C1

work function energy = hf_0

$= 6.63 \times 10^{-34} \times 1.00 \times 10^{15}$ C1

$= 6.63 \times 10^{-19}$ J A1

(ii) straight line with same gradient M1

displaced to right A1

(iii) For the same incident frequency and work function, B1

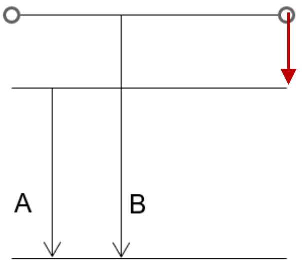
intensity determines number of photons arriving per unit time but

not the energy of the photon OR B1

it affects number of electrons emitted per unit time

and not maximum kinetic energy of electrons.

(b) **(i)** B1 [1]



(ii) $\Delta E = \frac{hc}{\lambda}$

EB = hc/488 x 10^{-9} = 4.07 x 10^{-19}
EA = hc/3.04 x 10^{-9} = 3.04 x 10^{-19} C1
ΔE = 1.03 x 10^{-19} C1

$\lambda = \frac{hc}{\Delta E} = 1.9 \times 10^{-6}$ m A1 [3]

(c) (i) Use $\Delta p\, \Delta x \geq h$
Δp = 6.63 x 10^{-34}/ 1.00 x 10^{-10} = 6.63×10^{-24} kg m s^{-1} for electron C1

%Δp/p = (6.63×10^{-24}/9.11×10^{-31} x 3.00 × 10^{7}) x 100 % = 24.3 % A1

7(ai)	u_y = 20 sin 50° = 15.3	[B1]

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	taking upwards as positive, $s_y = u_y\, t + ½\, gt^2$ $-52 = 20 \sin 50^o\, t + ½(-9.1)t^2$ $-52 = 15.3\, t – 4.91\, t^2$ $t = \dfrac{15.3 \pm \sqrt{15.3^2 - 4 \times 4.91 \times (-52.0)}}{2 \times 4.91}$ $= 5.17$ s - First B1: vertical component correctly resolved - C1: use of equation for vertical direction. - Second B1: correct substition	[C1] [B1]
(aii)	Horizontal displacement = 5.17 x 20 cos 50 = 66.5 m	[M1] [A1]
(aiii)	vy = uy + at $v_y = 15.3 + (-9.81)(5.17)$ = -35 .42 = - 35.4	[M1] [A1]

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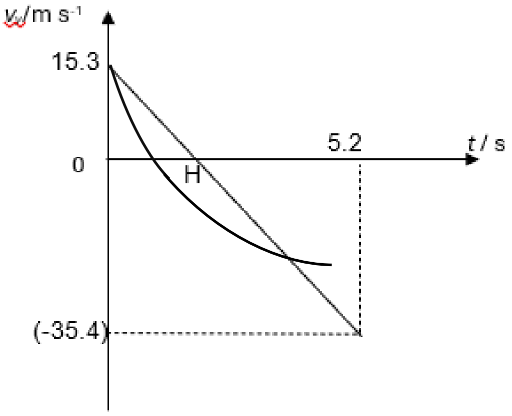
(aiv)	 B1- Straight line with negative gradient B1- correct labels of H and values of vertical velocities (with ECF), area under graph of first triangle smaller.	
(av)	B1: curve with gradient of curve before H steeper than (iv), max height reached at shorter time. B1: curve with gradient less steep than graph of (iv) and area under graph smaller	[B1] [B1]
(bi)	The momentum of a system of objects remains constant if no resultant external force acts on the system.	[B1]
(bii)	0 = (- (1550 – m) 0.45) + m (20 cos 50) 697.5 – 0.45 m = 12.86 m 697.5 =13.31 m m = 52 kg	[M1] [A1]
(biii)	Explanation or evidence of working of total KE before and after. Total KE before firing = 0 Final KE after firing = ½ (52)$(20)^2$ + ½ (1550 -52)$(0.45)^2$ Total KE not conserved Not elastic.	[B1] [B1]
(biv)	On the ball ; <F>t = mΔv <F> = (52) (20-0)/0.35 = 2971 N = 3000 N By Newton's Third Law Force on ball = force on cannon (statement required, otherwise minus 1 mark) = 3000 N	[M1] [M1] [A1]

(c)	No, the vertical momentum of the cannon is transferred to the ground vertically. / there is external force present.	[B1]

8	(a) (i)	$KE_i + PE_i = KE_f + PE_f$ $7.7 \times 10^{-13} + 0 = 0 + \frac{Qq}{4\pi\varepsilon_o r}$ $7.7 \times 10^{-13} = \frac{(79 \times 1.60 \times 10^{-19})(2 \times 1.60 \times 10^{-19})}{4\pi(8.85 \times 10^{-12})r}$ $r = 4.72 \times 10^{-14}$ m Since *r* is the distance of closest approach, the radius of gold must be less than this. Or Loss in KE = Gain in Electric PE	M1 M1 A0
	(a) (ii)	Top particle: smaller deviation (not zero deviation) Bottom particle: bigger deviation 	M1 M1
	(b) (i)	The binding energy per nucleon of the helium nucleus is defined as the average energy per nucleon needed to separate completely the nucleus into its individual 2 protons and 2 neutrons.	B1 B1
	(b) (ii)	BE = (mass defect) c^2 (BE in MeV) x 10^6 x *e* = ($2m_p + 2m_n - m_{helium}$) x *u* x c^2 4 x 7.07470 x 10^6 x *e* = [2 (1.00728 + 1.00866) – m_{helium})] x *u* x c^2 m_{helium} = 4.00157 u	C1 A1
	(b) (iii)	Total B.E of reactants = 14(7.47724) + 4(7.07470) = 132.980 MeV Total B.E of products = 131.788 MeV Since total B.E of reactants is greater than total B.E of products, this implies energy needs to be supplied for reaction to happen or Comparison of total mass. i.e. Mass of products > Mass of reactants Reaction cannot be spontaneous.	B1 M1 A1

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	(c) (i)	To deal with the random nature of the count rate, <u>draw a smooth curve to even out the fluctuations</u>.	B1
	(c) (ii)	To deal with the background radiation, at larger timings where the curve tapers off, <u>draw a horizontal line to obtain the background count rate</u>. Then deduct this background count rate from the observed count rate to obtain the actual count rate.	B1
	(c) (iii)	From the graph, background CR ~ 10.0 min^{-1}. CR on graph = 170 min^{-1} , CR due to source only = 170 – 10 = 160 min^{-1}. After $t_{1/2}$, CR on graph CR due to source only = 80 min^{-1}, CR on graph = 80 + 10 = 90 min^{-1} , For CR on graph to reduce from 170 to 90 min^{-1} $< t_{1/2} > \sim 1.6$ hours	B1 A1
	(c) (iv) 1.	half-life: no change because decay is spontaneous which means it is independent of environment and external factors like temperature.	B1
	(c) (iv) 2.	likely to be different as radioactive decay is random (and cannot be predicted).	B1
	(d)	A = $(3.8 \times 10^4)\, e^{(-\ln 2\,/\,15)(36)}$ = 7200 Bq $A \propto N \propto \text{volume}$ $\frac{A'}{A} = \frac{V'}{V}$ $\frac{1.2}{7200} = \frac{5}{V}$ $V = 3.0 \times 10^4 \text{ m}^3$	C1 C1 A1

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